

Experiment - 6

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Q. Create a Neural network (ANN,CNN) using the python programming language.

Aim/Overview of the practical:

To build and train an Artificial Neural Network (ANN) using TensorFlow for classifying the Iris dataset. The practical demonstrates data preprocessing, network training, and evaluation of model accuracy.

Task to be Done:

- ✓ Load a real-world dataset (Iris) and preprocess it (normalization and one-hot encoding).
- ✓ Design and construct an Artificial Neural Network (ANN) using TensorFlow.
- ✓ Compile the model with appropriate **loss function** and **optimizer**.
- ✓ Train the ANN on the training dataset and monitor learning progress.
- ✓ Evaluate the model on test data and report **accuracy**.

Code for Experiment

```
import tensorflow as tf
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler

data = load_iris()
X = data.data
y = data.target.reshape(-1, 1)

encoder = OneHotEncoder(sparse_output=False)
y_encoded = encoder.fit_transform(y)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_encoded, test_size=0.2, random_state=42)

model = tf.keras.Sequential([
    tf.keras.layers.Dense(10, activation='relu', input_shape=(4,)),
    tf.keras.layers.Dense(8, activation='relu'),
    tf.keras.layers.Dense(3, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Callback to print every 10 epochs
class PrintEvery10(tf.keras.callbacks.Callback):
    def on_epoch_end(self, epoch, logs=None):
        if (epoch + 1) % 10 == 0:
            print(f'Epoch {epoch+1}, Loss: {logs["loss"]:.4f}, Accuracy: {logs["accuracy"]:.4f}')

history = model.fit(X_train, y_train, epochs=50, batch_size=8, verbose=0,
                    callbacks=[PrintEvery10()])

loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print(f'\nTest Accuracy: {accuracy * 100:.2f}%')
```

Output :-

```
⚡ /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input_shape` / `input_dim`  
    super().__init__(activity_regularizer=activity_regularizer, **kwargs)  
Epoch 10, Loss: 0.7031, Accuracy: 0.7333  
Epoch 20, Loss: 0.5466, Accuracy: 0.7917  
Epoch 30, Loss: 0.4303, Accuracy: 0.8167  
Epoch 40, Loss: 0.3110, Accuracy: 0.8917  
Epoch 50, Loss: 0.2054, Accuracy: 0.9500  
  
Test Accuracy: 96.67%
```

Learning Outcomes (What I Have Learnt):

- ✓ Understand how to **build and train an Artificial Neural Network (ANN)** using TensorFlow.
- ✓ Learn to **prepare and preprocess real-world data** for neural network training.
- ✓ Gain knowledge of **activation functions and network architecture** for multi-class classification.
- ✓ Learn to **compile and optimize the model** using loss functions and optimizers.
- ✓ Evaluate model performance and interpret **accuracy on unseen test data**.